Network MCP Server with a Diagnosis Agent

An AI helper that runs real network tests and explains what is broken

21CSC302J Computer Networks · School of Computing, SRM IST

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Abstract

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When a network stops working, users only see the symptom, while the cause can sit at any layer: a weak signal, a wrong IP address or gateway, a broken route, or a DNS server that does not answer. Finding it needs the same step-by-step tests a network engineer runs.

This project builds a **Network MCP server** that exposes standard diagnostic tools (ping, traceroute, nslookup, port checks, interface, ARP and routing tables) through the Model Context Protocol, and a **diagnosis agent** that calls them from the bottom layer up, stops at the first failure, and explains the cause and the fix in plain language.

The design applies all five units of the course and is demonstrated on a Cisco Packet Tracer network with four faults, one per layer.

In one line

An AI helper that runs real network tests and tells you which layer is broken.

Problem
Symptoms hide the real layer

Approach
MCP tools + a bottom-up agent

Outcome
Cause and fix in plain words

Keywords: network diagnosis, MCP, AI agent, TCP/IP layers, troubleshooting

"My internet is not working"

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WHAT THE USER SEES

"YouTube won't load."

"Wi-Fi shows connected but no internet."

"Only one website is down?"

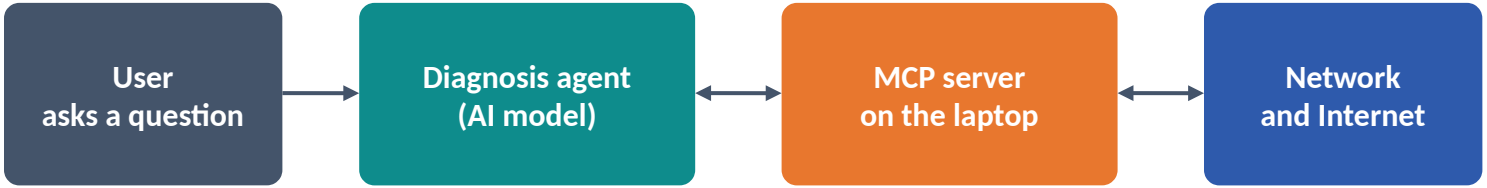
The real cause could be any one of these:

- Cable unplugged, weak Wi-Fi signal — Physical layer
- No IP address, wrong gateway — Network layer
- DNS not answering, port blocked — Transport and application

Our idea: an AI agent that runs real network tests on the computer, finds the layer that failed, and explains the fix in plain English.

How the system is built

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Tools the MCP server gives the agent

`ping`
Is the host reachable?

`traceroute`
Where does the path stop?

`nslookup`
Does the name resolve?

`port_check`
Is the service port open?

`ip / ifconfig`
Is the interface up, what IP?

`arp, route`
Neighbours and routing table

MCP = Model Context Protocol: a standard plug that lets an AI model call tools safely.

Diagnosing layer by layer

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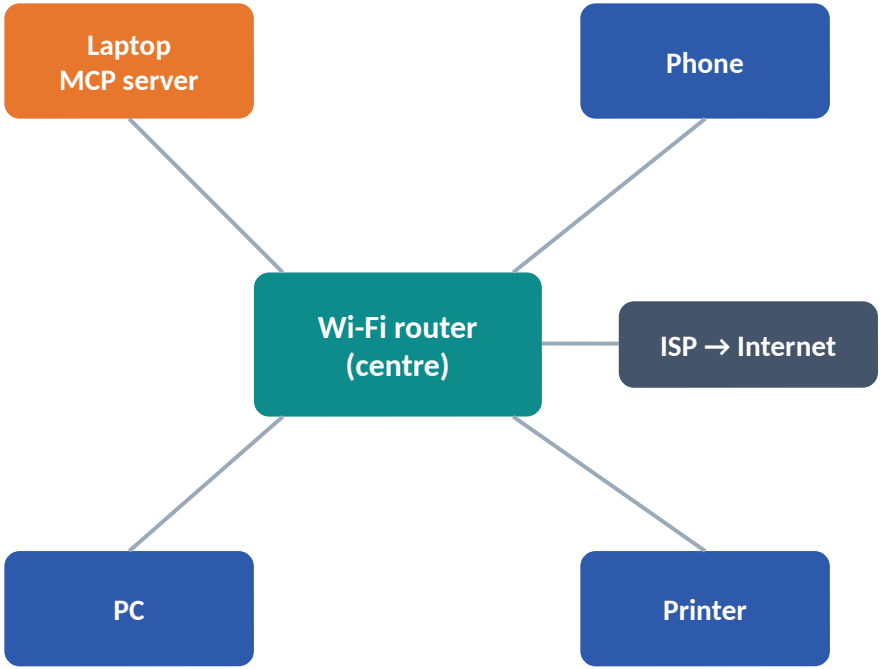
TCP/IP LAYER	WHAT LIVES THERE	AGENT CHECKS WITH
Application	DNS, HTTP, FTP	`nslookup, HTTP check`
Transport	TCP, UDP, ports	`port_check`
Network	IP, routing	`ping, traceroute, route`
Data link	MAC, ARP, Ethernet	`arp, error counters`
Physical	Cables, Wi-Fi signal	`interface status, signal`

Test from the bottom up. Stop at the first failure.

Network types and our test topology

Charith Reddy B

Type	Covers	Example
PAN	About 10 m	Phone to earbuds
LAN	Room or campus	Home Wi-Fi, college lab
MAN	A city	Bank branches in Chennai
WAN	Country or world	The Internet



Star topology: easy to set up, but if the centre fails, every device loses the network.

Physical layer checks

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Twisted pair
Guided · copper · LAN cables (Cat5e, Cat6)

Coaxial
Guided · copper · cable TV, old Ethernet

Fibre optic
Guided · light · fastest, long distance

Radio, microwave, infrared
Unguided · through the air · Wi-Fi, mobile

What the agent checks

- Is the interface up or down?
- Wired: link speed (100 Mbps, 1 Gbps)
- Wi-Fi: signal strength in dBm

Wi-Fi signal	Meaning
−30 to −60 dBm	Strong, fine for anything
−60 to −70 dBm	OK, may slow down
Below −75 dBm	Weak, drops and buffering

Latency, bandwidth and delay

Charith Reddy B

Latency — Time for data to travel. Ping shows round-trip time (RTT) in ms.

Bandwidth — How much data per second, e.g. 100 Mbps. Like the width of a road.

Jitter — How much the delay changes. Bad for calls and games.

Packet loss — Packets that never come back. Even 2 to 5% feels broken.

```
$ ping -c 4 8.8.8.8
64 bytes: time=21.4 ms
64 bytes: time=23.0 ms
64 bytes: time=19.8 ms
64 bytes: time=88.6 ms
4 sent, 4 received, 0% loss
avg 38.2 ms, jitter high
```

Agent says: reachable with no loss, but one slow reply. Usually Wi-Fi congestion, not a broken link.

IPv4 addressing: the first check

Rupesh Reddy

192.168.1.23 / 24

Network part
192.168.1

Host part
.23

/24 means the first 24 bits (mask 255.255.255.0) name the network.

Class	First number	Private range
A	1 to 126	10.0.0.0/8
B	128 to 191	172.16.0.0/12
C	192 to 223	192.168.0.0/16

Agent checks

1. **Has an IP address?**
   No address means no network at all.

2. **Starts with 169.254?**
   DHCP failed, the router never answered.

3. **Gateway set?**
   Without a gateway nothing leaves the LAN.

Subnetting: same network or not?

Rupesh Reddy

	/24	/26
Mask	255.255.255.0	255.255.255.192
Networks	1	4
Usable hosts	254	62 each
Example range	192.168.1.1 to .254	192.168.1.1 to .62
Broadcast	192.168.1.255	192.168.1.63

FLSM: every subnet the same size.
VLSM: sizes fit the need, less waste.

Agent's same-subnet check

```
my IP      192.168.1.23 /26
gateway    192.168.1.100

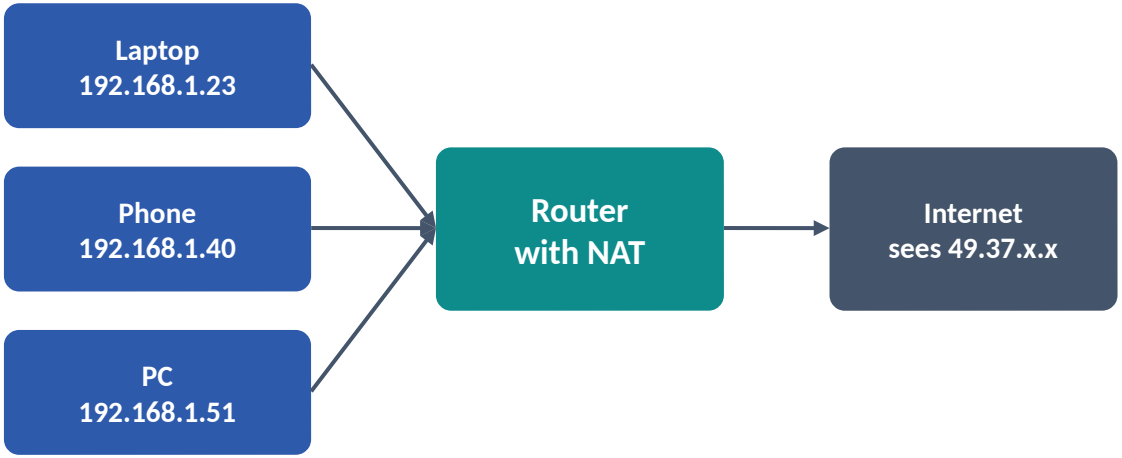
network    192.168.1.0
last host 192.168.1.62

gateway outside subnet ✗
```

Fix: set the gateway to an address inside 192.168.1.1 to .62, or widen the mask.

NAT and network devices

Rupesh Reddy



Device	Layer	Job
Hub	Physical	Repeats to every port
Repeater	Physical	Boosts a weak signal
Switch	Data link	Sends by MAC address
Bridge	Data link	Joins two LAN parts
Router	Network	Sends by IP, joins networks

Many private addresses share one public address.

Routing: how packets find the way

Basith Nistar

```
$ netstat -rn
Destination     Gateway         Iface
0.0.0.0/0       192.168.1.1     wlan0
192.168.1.0/24  directly        wlan0
127.0.0.0/8     loopback        lo
```

No 0.0.0.0/0 line means no way out of the LAN.

	Static routing	Dynamic routing
Set by	Admin, by hand	Protocol, automatically
Changes	Does not adapt	Adapts to failures
Good for	Small networks	Large networks

Routing protocols in our syllabus

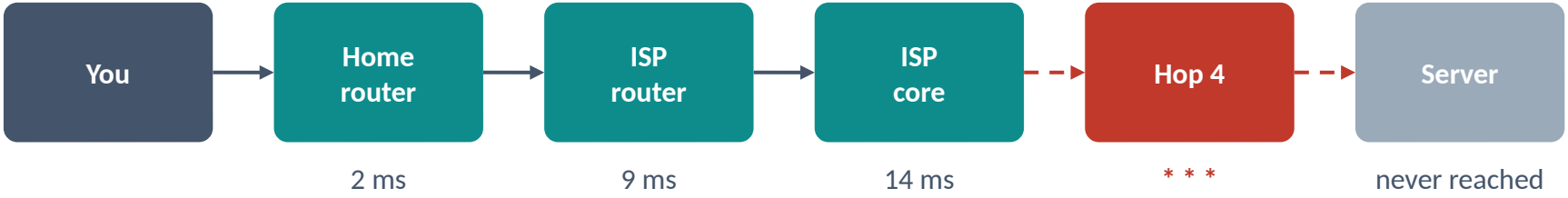
RIP — Distance vector. Picks the path with the fewest hops (max 15).

OSPF — Link state. Every router knows the map, picks the lowest cost.

BGP — Path vector. Connects ISPs and big networks to each other.

Traceroute: where the path breaks

Basith Nistar



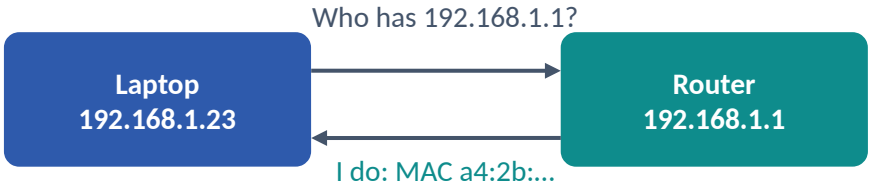

```
$ traceroute youtube.com
 1  192.168.1.1     2 ms
 2  10.21.0.1       9 ms
 3  103.21.44.9    14 ms
 4  * * *
 5  * * *
```

How it works

- Each packet has a TTL (time to live).
- Every router lowers TTL by 1; at 0 it sends back an error.
- TTL 1, 2, 3 … reveals each hop in order.
- Agent: path breaks after hop 3, inside the ISP.

Data link layer: MAC, ARP and errors

Basith Nistar



ARP turns an IP address into a MAC address.

Error check	How it works	Catches
Parity bit	One extra bit, even or odd	Single-bit errors
Checksum	Add up the data, send the sum	Most errors
CRC	Polynomial division remainder	Burst errors, best

Agent reads error counters

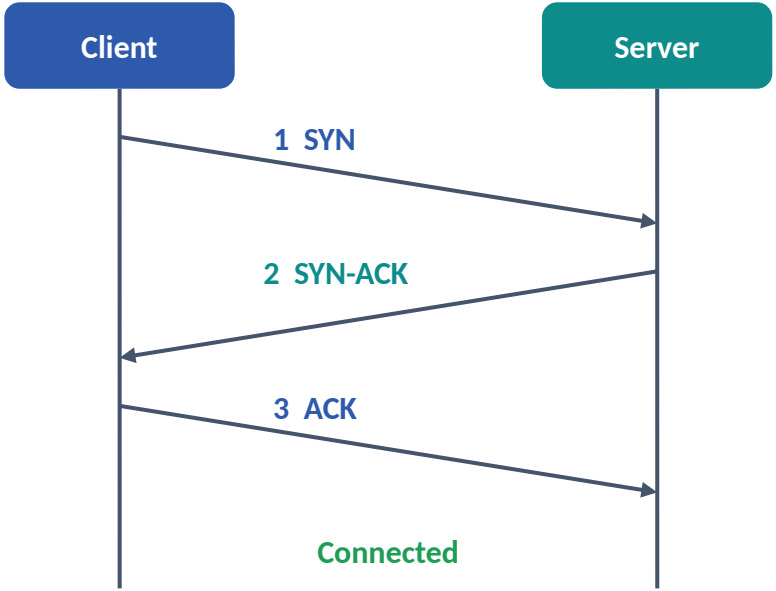
```
$ ip -s link show wlan0
RX: packets 182044
    errors 0  dropped 3
TX: packets 99710
    errors 0  dropped 0
```

Growing CRC or frame errors: likely a damaged cable or radio interference, a physical fault.

Transport layer: ports, TCP and UDP

Adhityan

TCP three-way handshake

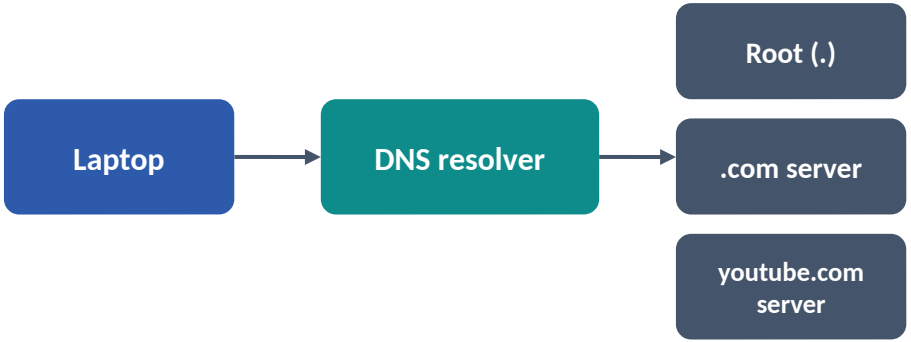


	TCP	UDP
Connection	Handshake first	None, just send
Reliable?	Yes, resends lost data	No
Order	Kept in order	Not guaranteed
Used by	Web, email, FTP	DNS, video calls, games

`port_check youtube.com 443` → handshake completes, so HTTPS is reachable.

Application layer: DNS and HTTP

Adhityan



1 Laptop asks the resolver for youtube.com
2 Resolver asks root, then .com, then youtube.com
3 Answer 142.250.x.x comes back and is cached

```
$ nslookup youtube.com
Address: 142.250.183.14
```

Service	Protocol	Port
DNS	UDP	53
HTTP	TCP	80
HTTPS	TCP	443
FTP	TCP	20, 21
Email (SMTP)	TCP	25
Telnet	TCP	23

Agent checks DNS first, then fetches the page over HTTP.

A diagnosis, start to finish

Adhityan

Question: "YouTube is not loading."

1	Interface up?	`wlan0 up, signal −58 dBm`	PASS
2	Got an IP?	`192.168.1.23/24`	PASS
3	Reach the gateway?	`ping 192.168.1.1 → 2 ms`	PASS
4	Reach the Internet?	`ping 8.8.8.8 → 21 ms`	PASS
5	Name lookup?	`nslookup → timed out`	FAIL

Example run to show the flow.

AGENT'S ANSWER

Your Internet works, but the DNS server is not answering, so names like youtube.com cannot be found.

Fix: set DNS to 8.8.8.8 or 1.1.1.1.

Summary

One project, every unit of the syllabus

Unit 1	Network types, topology, media, performance
Unit 2	IPv4, subnetting, NAT, devices
Unit 3	Routing tables, RIP, OSPF, BGP, traceroute
Unit 4	MAC, ARP, error detection
Unit 5	TCP, UDP, ports, DNS, HTTP

Future scope

- Fix problems after asking the user
- IPv6 checks
- Run against Packet Tracer labs

Thank you. Questions?